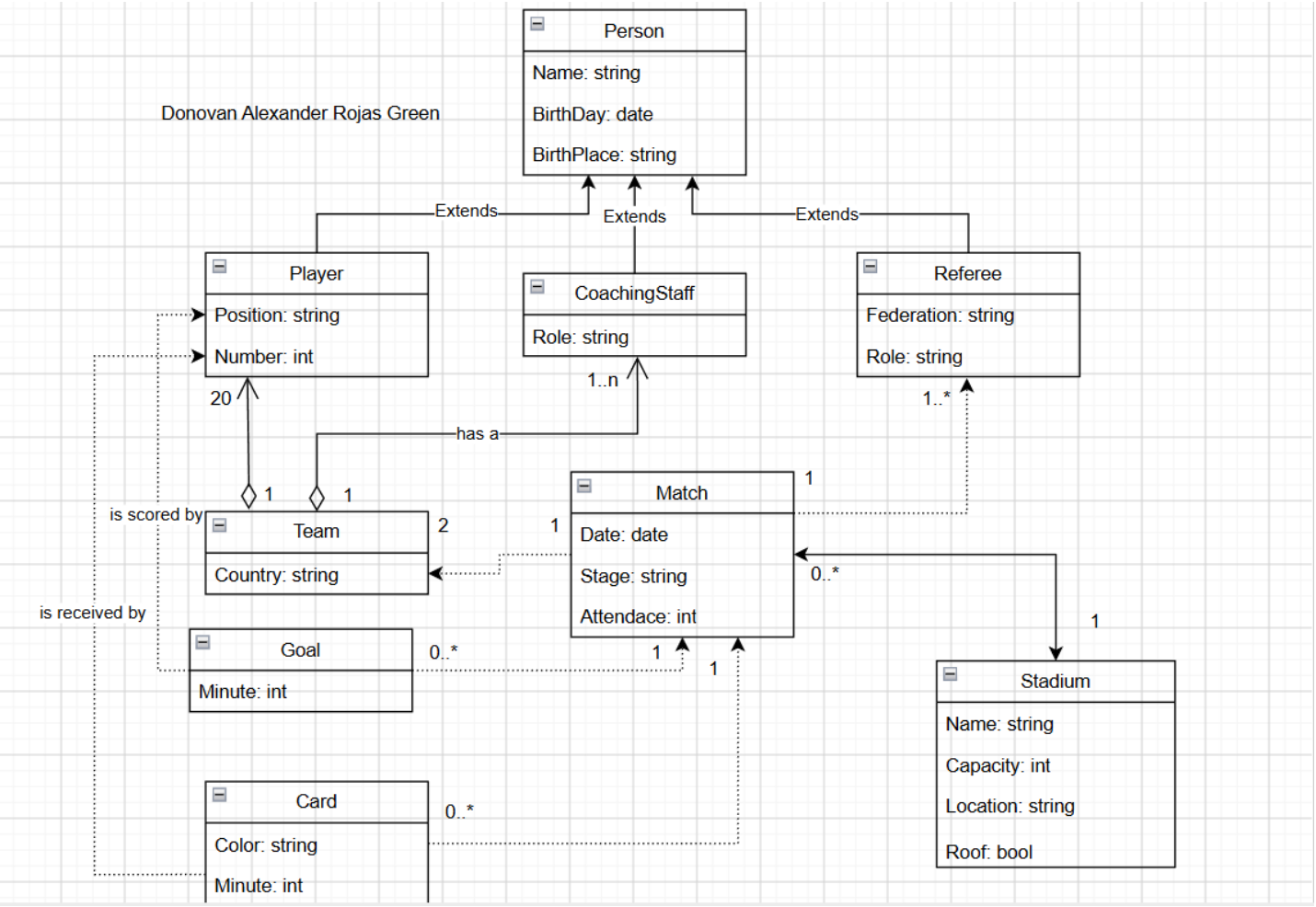
# Justification of the Class Diagram

**OLYMPIC FOOTBALL COMPETITIONS**

**DONOVAN ALEXANDER ROJAS GREEN**

## 1. CLASS DIAGRAM



# 2. The problem being modeled

## Class diagram exercise

- The International Olympic Committee (IOC) wants to analyze, design, and implement a solution proposal that allows the committee to track all details related to the Olympic football competitions:

- The football competitions will take place across a number of stadiums.

- Each match will involve two teams and a designated group of referees.

- For each match, the system must capture data such as goals scored, cards issued, spectator attendance, and the specific stage of the tournament.

- Each team consists of 20 players and a coaching staff.

- Required information for each player includes their name, date and place of birth, playing position, and number.

- For the coaching staff, the necessary details include name, date and place of birth, and role.

- For referees, the required information includes name, date and place of birth, affiliated federation, and role.

- Regarding goals and cards, the system needs to record the match, the minute of the event, and the person involved.

- The organizing committee wishes to track basic stadium characteristics, such as name, capacity, location (city), and whether the stadium has a roof or not.

## 3. Why a class diagram was chosen

A class diagram is used to show the static structure of a system—that is, what "things" (entities) exist, what data they store, and how they are logically connected.

It is the right tool here because the requirement focuses on storing and relating information (data regarding players, stadiums, and match events).

A sequence diagram shows the temporal order of a process (such as making a reservation) but does not detail the attributes of each object. In this case, we need to clearly see the characteristics of each element and their multiplicities (how many players per team, how many teams per match), making the class diagram indispensable.

## 4. The classes (participants): why these and not others

• Person (Superclass): This class was created to group common attributes (Name, BirthDay, BirthPlace). Without it, we would have to repeat these three data points in Player, CoachingStaff, and Referee, which is redundant and poor practice.

• Player / CoachingStaff / Referee: These are specific classes that inherit from Person. They were separated because each has unique data (e.g., a player has a position and number; a referee belongs to a federation).

• Team: Groups players and coaching staff together. It serves as the necessary link allowing a group of people to participate in a match.

• Match: The system's central class. It connects the competing teams, the officiating referees, and the stadium where the event takes place.

• Stadium: Modeled as a separate object because it possesses its own state and characteristics (capacity, roof) regardless of whether a match is taking place.

• Goal / Card: Modeled as independent classes rather than simple attributes within Match. This is because a match can involve multiple goals or cards, and the system needs to track exactly when they occurred (Minute) and which Player was involved ("is scored by" / "is received by").

## 5. Overview and justification of each relationship

| Element / Relationship | Justification (why it exists) | Notation used |
|---|---|---|
| **Inheritance** (*Person* to *Player*, *Coach*, *Referee*) | The requirement asks for identical data for these three roles (name, birth details). They are centralized in *Person*. | Solid line with an empty arrowhead (Extends) pointing to the parent class. |
| **Composition** (*Team* to *Player*) | The prompt is strict: "Each team consists of 20 players". The player roster is a vital and inseparable part of the team for the tournament. | Line with a black diamond (composition) on the *Team* side and a multiplicity of `20`. |
| **Relationship / Aggregation** (*Team* to *CoachingStaff*) | Each team needs a coaching staff. | Line with a diamond with multiplicity `1` to `1..n` (at least one staff member). |

| Element / Relationship | Justification (why it exists) | Notation used |
|---|---|---|
| **Association** (*Match* to *Team*) | A match, by definition in the requirement, involves exactly two competing teams. | Dashed line with multiplicity `1` (*Match*) to `2` (*Team*). |
| **Association** (*Match* to *Referee*) | A match requires a designated group of referees. | Dashed line from *Match* to *Referee* with multiplicity `1..n` (one or many). |
| **Association** (*Match* to *Stadium*) | Competitions take place in stadiums. A stadium can host many matches (`0..n`), but a specific match is played in only one stadium (`1`). | Solid line with an arrow, multiplicity `0..n` to `1`. |
| **Dependency/Association** (*Match* to *Goal* / *Card*) | The system registers cards and goals per match. | Dashed lines indicating that these events belong to a specific *Match*. |
| **Action relationship** (*Goal*/*Card* to *Player*) | It is necessary to record "the person involved" in the event. The diagram specifically indicates which player scores ("is scored by") or receives the card ("is received by"). | Dashed line with descriptive text of the association's role. |

## 6. CONCLUSION

If dividing problem into parts is simple of answer, first I have the class person with attributes name. birthday, birthplace; the name and birthplace declared like string in case they are alphanumeric, birthday declared like

date after the class player with attributes postion like string and number like int, coachingstaff with attributes role declared like string and the class referee with attribute federation like string and role like string, these three class depend on the class person since more apart from being player, coachingstaff or referee are person, the class team depends on class player and coachingstaff since no player or coachingstaff dont exist team, without teams or referee there is no match, the goals and card depends on player and also on the referees but in the diagram it is not shown and without stadium there is no match. Regarding multiplicity, the text indicates that a team consists of twenty players; therefore, the relationship is twenty players to one team. There is also a one-to-n relationship between the team and the coaching staff, meaning a team can have one or more coaching staff members. Between the match and team classes, there are two teams but only one match. Regarding the match and referee classes, a match requires a referee, resulting in a one-to-one relationship. For goals and matches, there can be zero goals even if a match takes place—the same applies to cards—hence the 0-to-1 notation; similarly, a stadium can exist without hosting any matches, resulting in a 0-to-1 relationship.